International study of 24-hour movement behaviours among preschool children (SUNRISE): A pilot study from Nepal

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Resumo (Abstract)

Background This study aimed to determine the proportion of Nepali preschool children (3–4 years of age) meeting the World Health Organization (WHO) 24- hour movement guidelines for physical activity (PA), sedentary behaviour and sleep by sex (male vs. female) and place of residence (urban vs. rural) and evaluate their executive functions and motor skills. Furthermore, it evaluated the feasibility and acceptability of methods proposed for the main SUNRISE study in diverse settings. Methods In this pilot study, a total of 99 preschool children, balancing sex and place of residence, each with one parent/caregiver, from Syangja District were recruited. Physical activity, sedentary behaviour and sleep were assessed using waist-worn ActiGraph accelerometers (ActiGraph wGT3x-BT) with ≥ 1 valid 24-hour period of data, motor skills (gross and fine) were evaluated using the NIH Toolbox motor battery, and executive functions were assessed using the Early Years Toolbox. Eight focus group discussions (FGDs) were carried out with parents/caregivers, and a short interview was conducted with eight Early Childhood Development (ECD) centre teachers to explore the feasibility and acceptability of implementing the SUNRISE protocol. Results Only 10% of children met all components of the WHO 24-hour movement guidelines (≥ 180 min total PA, of which ≥ 60 min were MVPA, ≤ 1 h of screen time, and 10–13 h of sleep). Proportions of those meeting the guidelines were higher for sleep (81%) and physical activity (79%) than for screen time (56%). Altogether, 96 children wore the accelerometer, of which 78 had at least ≥ 1 valid 24-hour period of data. Males and rural children accumulated more daily moderate-to-vigorous and vigorous physical activity than females and urban children, respectively. In contrast, females were more likely to meet screen time guidelines than males (66% vs. 42%, p = 0.018). Additionally, females had better inhibition control than males, and rural children had better working memory than urban children. Parent/caregivers and ECD teachers reported that the study procedures, including accelerometer use, were acceptable and feasible. Conclusions Adherence to the full set of WHO movement guidelines was very low among preschool children in this study, with screen time emerging as the most challenging component. Targeted strategies are needed to promote healthier movement behaviours, particularly to reduce screen time. The findings support the feasibility and acceptability of conducting the main SUNRISE study in Nepal. Supplementary Information The online version contains supplementary material available at https://doi.org/10.1186/s13690-026-01984-2 .

Background

Childhood overweight and obesity remain major global public health challenges in the 21st century, with profound implications across the life course [1]. Excess adiposity in early childhood tracks into adolescence and adulthood, substantially elevating the risk of non-communicable diseases (NCDs), including type 2 diabetes and cardiovascular disease [1 – 3]. According to the World Health Organization (WHO), in 2024, approximately 38 million children under five years of age were living with overweight or obesity globally, with nearly half residing in Asia [1]. Among adolescents aged 15–19 years, the prevalence of overweight and obesity has increased more than fourfold between 1990 and 2022, reaching approximately 390 million in 2022 [1]. These trends underscore the urgent

need for early-life prevention strategies, particularly in low- and middle-income countries, such as Nepal, which are undergoing a rapid epidemiological transition [4]. In Nepal, the prevalence of overweight and obesity among children under five years remains relatively low (1.4%) [5]. However, substantially higher prevalence rates, up to 27%, have been reported among school-aged children (6–13 years), particularly in urban settings [6 , 7]. These emerging patterns suggest a shift towards obesogenic environments, especially in rapidly growing and often unplanned urban areas [8]. This trend is concerning, as childhood obesity is strongly associated with hypertension, insulin resistance, early markers of cardiovascular disease, psychological disorders, and an elevated risk of premature mortality, disability, and persistent obesity in childhood, adolescence, and adulthood [1]. Preventive efforts during the early years are therefore critical.

Movement behaviours, including physical activity, sedentary behaviour (such as screen time), and sleep, are key modifiable determinants of healthy growth and development and play a central role in the prevention of childhood overweight and obesity [9 , 10]. Importantly, these behaviours are codependent within 24 h; time allocated to one behaviour necessarily displaces time from another [11].

Recognising this interrelationship, integrated 24-hour movement guidelines have been advocated to promote a holistic approach to early childhood health. Following the introduction of Canada and Australia's 24-Hour Movement Guidelines in 2017 [12 , 13], WHO released guidelines on physical activity (at least 180 min each day, including at least 60 min of energetic play), sedentary behaviour (no more than 1 h of screen time per day), and sleep (10–13 h of good quality sleep each night) for children under five years in 2019 [14]. These guidelines provide an evidence-informed framework to support public health action globally.

However, to inform implementation, an assessment of adherence to these recommendations was needed across diverse settings. The International Study of Movement Behaviours in the Early Years (SUNRISE) [15] was established to evaluate compliance with the WHO guidelines and to examine associations between 24-hour movement behaviours and health, and assess developmental and cognitive outcomes across countries of varying income levels [15]. Before full-scale implementation, a pilot study is necessary to assess the feasibility of the study protocol [15]. In this context, we conducted the SUNRISE pilot study in Nepal to determine the proportion of 3-4-year-old children meeting the WHO movement guidelines, examine executive functions and motor skills, and evaluate the feasibility of implementing the SUNRISE protocol in urban and rural contexts. This paper presents the SUNRISE pilot study Nepal findings.

Methods

Study design and participants

We carried out a cross-sectional survey in two municipalities of Syangja District in Gandaki Province, located in western Nepal. Syangja is a hilly district consisting of 11 municipalities (5 urban and 6 rural). We purposively selected Waling Municipality and Biruwa Rural Municipality to represent urban and rural settings, respectively. In Nepal, urban areas are defined as settlements officially designated as municipalities, generally characterised by a population exceeding 5,000, a population density of at least 10 persons per hectare, and at least 50% of the population engaged in non-agricultural activities. Settlements outside these municipalities are classified as rural areas [16 , 17]. The study participants included children attending preschool (hereafter referred to as Early Childhood Development [ECD] centres), one of their parents/caregivers, and ECD teachers.

Participant recruitment and data collection

First, we sought approval from the respective municipalities and obtained a list of schools operating ECD programs within their jurisdictions. From this exhaustive list, we purposively selected ECD centres with a large enrolment to ensure adequate representation. At this stage, three ECD centres were selected from Waling Municipality (urban) and five from Biruwa Rural Municipality (rural). Two field researchers with an undergraduate degree in Public Health visited all eight centres, submitted a request letter to the principal and asked for approval to enroll study participants. All the ECD centres consented to the study.

In total, we found 310 children (210 in the urban and 90 in the rural) across the eight centres. We then allocated the number of participants proportionate to the enrolment size of each centre to balance the required sample size of 100 children for the pilot study (50 each in urban and rural) [15]. In Waling Municipality, we selected 20 children from two ECD centres, and 10 from the third centre. In Biruwa Rural Municipality, we selected 10 children from each of the five ECD centres. We randomly selected participants stratified by sex to ensure a balanced representation of males and females in each centre. We reached out to the parents/caregivers by sending an invitation letter through their child and/or via telephone contact, supported by the ECD teacher.

Data were collected between July and September 2022, after the country had resumed normal operations following the COVID-19 pandemic. Eligible participants were children aged three to four years at the time of data collection. Parents/caregivers of eligible children were invited to attend the ECD centre in groups of up to 15, based on the number of accelerometers available. Parent/caregivers provided the written informed consent. All the data collection tools, including consent and assent forms, were translated into the Nepali language. Parent/caregivers and ECD teachers completed the questionnaires as per the SUNRISE study protocol on the same day [18]. Parent/caregiver questionnaires included sociodemographic information and their child's 24-hour movement behaviours. Similarly, an ECD teacher at each centre completed a questionnaire related to centre-based policies on 24-hour movement behaviours, children's nap times, their physical activity practices, and food provisions at the centre. All the parents/caregivers and ECD teachers completed the questionnaires.

The field researchers completed a two-day in-person training in the capital city, Kathmandu, organised by a trainer from the SUNRISE study coordinating centre at the University of Wollongong. The researchers collected the data in person at all study sites. They measured children's height and weight, assessed fine and gross motor skills, and executive functions within the ECD setting, following the standardised procedures. All measurement tasks (i.e., anthropometry, fine and gross motor skills, and executive function) were conducted sequentially in accordance with the SUNRISE study protocol (version 5.3) [15]. Following the assessment, researchers fitted each child with an ActiGraph accelerometer and instructed them to wear it for five consecutive days to get three full days (3 × 24-hour period) of data [18]. The field researchers collected the accelerometers from the respective ECD centres on the fifth day.

Measures

Anthropometrics

We measured the height and weight of the child using a portable wall stadiometer (SECA 213) and a digital scale (SECA 878), respectively. Children wore light clothes and no shoes during the measurement. Two measurements were taken, and the average was considered, rounding the value

to the nearest 0.1 cm or kg [18]. We calculated the body mass index (BMI) using the standard formula, i.e., weight in kilograms divided by height in meters squared (kg/m 2). Weight-for-age, height-for-age, weight-for-height, and BMI-for-age z-scores were then calculated using the WHO Child Growth reference standards [19].

Executive functions

We assessed the executive functions that included control of behavioural urges and impulses (inhibition), and visual-spatial working memory (memory) of the children using two validated iPad games (https://www.eytoolbox.com.au)(20) “GO/No-Go”, and “Mr Ant”, respectively. The “GO/No-Go” assessed the behavioural urges and impulses. In this game, fish and sharks were presented, and the child was instructed to tap the iPad screen when fish appeared and not to tap when sharks appeared. The total score was calculated as percentage accuracy (range between 0 and 1, with 1 referring to a perfect score) by multiplying the proportions of go trials (catching the fish) and no-go trials (not catching the shark) [20]. Mr. Ant game measured the child’s visual-spatial working memory, i.e., the amount of visual information that can be actively maintained. In this game, an ant was presented with several colored dots in different spatial locations on its body, which vanished after some time. Then, the child was asked to recall and tap the locations of the dots. The scoring begins at level 1, and participants earn 1 point for each consecutive level in which at least two of the three trials were accurate. After the initial levels, participants also earn an additional third of a point for each correct trial thereafter. Therefore, the final score could range between 0 and 8 points [20].

Gross and fine motor skills

The National Institute of Health (NIH) Toolbox (Motor battery) was used to assess a child’s fine and gross motor skills [21]. Each skill was measured twice using validated activities and scored according to the NIH Toolbox guidelines. In the SUNRISE study, five motor skill tests were included [18].

For the fine motor skill test, 1) the nine-hole pegboard test (PAT-A8515, Sammons Preston, Illinois, USA) was used to assess dexterity or manipulation. For gross motor skills, four tests were used: (1) a supine-timed up-and-go (S-TUG) – to assess mobility and posture; (2) a one-legged standing balance test – to measure posture and balance; (3) a standing long jump - to determine lower body explosive strength and mobility; and (4) a handgrip dynamometer (TKK5825, Grip-A, Takei, Tokyo) – to assess upper extremity strength [18].

Sleep, screen time, and restrained sitting

Sleep duration was calculated as the sum of daytime nap and nighttime sleep as reported by the accelerometer. Consistency of sleep and nap timings was confirmed with parents/caregivers and ECD teachers using a set of generic questionnaires [18 , 22]. They were asked to report their child’s typical bedtime routine, wake-up time, sleep quality, and nap times over 24 h. Example questions included: Does your child have a consistent bedtime/wake-up time? Similarly, ECD teachers were asked about the start and end times of naps during the children’s time at the centre. An example question was: What time does nap time start and end? (Please provide the time in hours and minutes using the 24-hour format). Screen time was defined as time spent using electronic devices such as smartphones, watching television, or playing video games on tablets or computers. Parent/caregivers were asked to report how much time during 24 h their child spent using screens while sitting or lying down, as well as time spent in restrained sitting (e.g., strapped in or unable to move) over the past

week [15].

Accelerometry

We assessed children's physical activity, sedentary behaviour, and sleep patterns using the ActiGraph wGT3x-BT tri-axial accelerometer (ActiGraph Pensacola, FL, USA) worn on the right hip. Each child wore the device for five consecutive days, with the ActiGraph removed only for water-based activities. ActiGraph data were collected at 30 Hz and processed into 15-second epochs using ActiLife software. A valid dataset required at least one 24-hour period with a minimum of six hours of wear time during waking hours, confirmed by visual inspection [23]. Periods with 20 or more consecutive minutes of zero counts between 5 am and 11:30 pm were considered non-wear time and excluded. Data were filtered using a standardised wake–sleep window (7:20 AM–9:10 PM), and physical activity intensities were categorised using Pate cut-points, i.e., sedentary behaviour [< 800 counts per minute (CPM)], light intensity PA [(LIPA) 800–1679 CPM], moderate-intensity PA [(MPA) 1680–3367 CPM], and vigorous-intensity PA [(VPA) (≥ 3368 CPM)] [24]. Total PA (TPA) was objectively measured and quantified as total daily movement time, defined as the sum of light, moderate and vigorous physical activity [14]. Bedtime and wake-time were determined by combining inclinometer data with parent/caregiver and ECD teacher-reported sleep and nap times. Bedtime was defined as the first minute of 10 consecutive minutes with zero vector magnitude, occurring at least 90 min before wake-time. Wake-time was defined as 10 consecutive minutes with a vector magnitude greater than zero [25 , 26].

Meeting the 24-hour movement guidelines

We classified the preschool aged children meeting the guidelines as per the WHO Global recommendations, that includes: (a) at least 180 min of total physical activity (TPA) of which 60 min are moderate to vigorous physical activity (MVPA) per day; (b) screen time no more than 1 h per day; and (c) sleep duration of 10–13 h (including naps) [14]. MVPA and TPA data were collected using accelerometers, whereas screen time and sleep/nap times (bedtime and wake-up time) were parent/caregiver and teacher-reported.

Acceptability and feasibility

To assess the acceptability of the measures used and the feasibility of implementing the study protocol at scale, we conducted eight Focus Group Discussions (FGDs) across the eight ECD centres, each involving 5–9 parent/caregiver participants. In addition, we interviewed eight ECD teachers, one per centre. Data were recorded in written form through paper-based questionnaires and field notes. Both the FGDs and interviews were brief, lasting approximately 10–20 min.

We used semi-structured questionnaires developed specifically for the SUNRISE pilot study and administered separately to parents/caregivers and ECD teachers. The parent/caregiver questionnaire addressed four key areas: comprehension of the questionnaire items, their child's engagement during data collection, feedback on their child wearing the accelerometer, and suggestions for improving data collection methods. Example questions included: "How did you find the parent/caregiver questionnaire?" and "Were there any questions that you did not understand or found difficult to answer?"

The ECD teacher questionnaire covered five key areas: teachers' observations during data collection, their perceptions of children's responses to wearing the accelerometer, children's engagement in the gross and fine motor skill assessments, and recommendations for future studies. Example questions

included: “How did you find the children wearing the accelerometer for five days?” and “Were there any problems? If so , could you describe them and provide suggestions for improvement?”

Data analysis

We used Stata v18 for statistical analysis. The descriptive statistics (mean, SD and frequency or percentage) were used to assess the proportion of preschool children meeting the WHO guidelines. The scores were calculated as meeting or not meeting the guidelines, separately and in combination with the recommendations [18]. We examined the differences by sex (males/females) and place of residence (urban/rural) using the Wilcoxon rank-sum test for continuous variables and the chi-square test for categorical variables. Statistical significance was assessed at a two-tailed alpha level of 0.05. Kernel density plots were generated to visualise the distribution of z-scores for the study sample in comparison with the WHO reference distribution.

The principal investigator (NS), who holds a doctoral degree in public health, thematically analysed the qualitative data from the FGDs and interviews with parent/caregivers and ECD teachers. Before coding, all transcripts were translated from Nepali into English. A deductive approach was employed to categorise the data into themes, as outlined in the SUNRISE study protocol [18]. The findings were organised under four main themes: parent/caregiver feedback on the questionnaire, children’s overall response to participation, feedback on accelerometer use, and feedback on executive and motor skill tasks. Results from the ECD teacher interviews are reported separately under the theme: ECD teachers’ observations, as per the SUNRISE study protocol [18].

Results

A total of 99 preschool children participated in the study, comprising 49 from urban and 50 from rural areas. One refused to participate in the study. The mean age was 3.8 years (SD = 0.6). The sex distribution was approximately equal (~ 50% males). Table 1 presents the anthropometric characteristics, time spent in different activity intensities, and movement behaviours stratified by sex and place of residence. We found no significant differences in anthropometry by sex of the child, but found significant sex differences in MVPA, VPA, and screen time (Table 1). Males had higher MVPA, VPA and screen time than females. When compared by place of residence, children from rural areas engaged in more minutes of physical activity (TPA, MVPA, and VPA) and screen time than their urban counterparts (Table 1). Rural children had lower BMI-for-age and weight-for-height z-scores than urban children.

Table 1 Descriptive characteristics [(Mean ± SD) or n (%)] of children and time spent on movement behaviours, stratified by sex and place of residence Characteristics All (n=99) Males (n=41) Females (n=58) p-value Urban (n=49) Rural (n=50) p-value Age (years) 3.8 ± 0.6 3.8 ± 0.6 3.8 ± 0.6 0.530 3.6 ± 0.4 4.0 ± 0.6 <0.001* Weight (kg) 14.2 ± 1.9 14.6 ± 1.8 14.0 ± 1.9 0.170 14.1 ± 1.9 14.4 ± 1.9 0.520 Height (cm) 96.9 ± 5.3 97.7 ± 5.6 96.4 ± 5.0 0.260 95.7 ± 5.2 98.2 ± 5.1 0.017* BAZ -0.2 ± 0.8 -0.2 ± 0.9 -0.2 ± 0.8 0.700 -0.0 ± 0.8 -0.4 ± 0.9 0.025* HAZ -1.1 ± 1.0 -1.0 ± 1.1 -1.2 ± 0.9 0.330 -1.1 ± 1.1 -1.1 ± 0.9 0.870 WAZ -0.8 ± 0.9 -0.7 ± 0.9 -0.9 ± 0.9 0.400 -0.7 ± 0.9 -1.0 ± 0.9 0.160 WHZ -0.3 ± 0.9 -0.3 ± 0.9 -0.3 ± 0.8 0.900 -0.1 ± 0.8 -0.5 ± 0.9 0.036* TPA (min/day) 1 192.8 ± 48.0 204.1 ± 50.1 185.0 ± 45.4 0.084 175.6 ± 38.7 205.4 ± 50.5 0.006* MVPA (min/day) 1 84.0 ± 28.5 93.3 ± 31.0 77.6 ± 25.0 0.016* 73.5 ± 24.0 91.7 ± 29.4 0.005* VPA (min/day) 1 16.0 ± 9.6 19.8 ± 10.2 13.4 ± 8.2 0.003* 12.9 ± 8.0 18.3 ± 10.1 0.014* Sedentary behaviour (min/day) 595.3 ± 57.3 582.2 ± 61.1 604.5 ± 53.3 0.090 606.6 ± 41.4 587.1 ± 65.9 0.140 Screen time (min/day) 86.3 ± 81.2 109.1 ± 97.9 70.1 ± 62.9 0.018* 69.0 ± 61.3 103.2 ± 94.4 0.035* Sleep (min/day) 629.2 ± 67.5 623.2 ± 52.6 633.4 ± 76.5

0.460 636.0 ± 75.8 622.5 ± 58.2 0.320 * denotes the value is significant, Abbreviations : n number of children, SD Standard deviation, BAZ Body mass index z-score, HAZ Height-for-age score, WAZ Weight-for-age score, WHZ Weight-for-height score, TPA Total physical activity, MVPA Moderate-to-vigorous physical activity, VPA Vigorous physical activity

The distributions of weight-for-age and height-for-age z-scores were shifted left compared with the WHO reference standards, indicating lower weight and linear growth relative to age (Supplementary Figs. 1 and 2). In contrast, weight-for-height and BMI-for-age z-score distributions largely overlapped with WHO references, suggesting broadly similar body proportionality despite lower size for age (Supplementary Figs. 3 and 4).

Table 2 presents the proportion of children meeting the WHO recommendations, both individually and in combination, across the five indicators. Accelerometry data were obtained from 93 children, but six data sets were lost during transfer/not retrieved. Of the remaining 87 participants, 78 had at least 24 h of valid data with a minimum of six hours of wear time during waking hours and were included in the final analysis. Among these 78 children, 5% each contributed one to two days of valid data, while the remaining 90% contributed more than two days.

Table 2 Children meeting each of the 24-h movement guidelines [n (%)], stratified by sex and place of residence Characteristics Total n (%) Males n (%) Females n (%) p -value Urban n (%) Rural n (%) p -value ≥ 60 min MVPA per day 1 62 (79.5%) 27 (84.4%) 35 (76.1%) 0.370 24 (72.7%) 38 (84.4%) 0.210 ≥ 180 min TPA per day 1 47 (60.3%) 21 (65.6%) 26 (56.5%) 0.420 16 (48.5%) 31 (68.9%) 0.069 ≤ 60 min screen time per day 2 55 (55.6%) 17 (41.5%) 38 (65.5%) 0.018* 32 (65.3%) 23 (46.0%) 0.053 10–13 h sleep per day 2 81 (81.8%) 34 (82.9%) 47 (81.0%) 0.810 39 (79.6%) 42 (84.0%) 0.570 < 1 h restrained sitting at a time 2 37 (37.4%) 13 (31.7%) 24 (41.4%) 0.330 17 (34.7%) 20 (40.0%) 0.590 Meeting all recommendations 1 8 (10.3%) 0 (0.0%) 8 (17.4%) 0.013* 3 (9.1%) 5 (11.1%) 0.770 Abbreviations : n Number of children, TPA Total physical activity, MVPA Moderate-to-vigorous physical activity, * denotes the value is significant Analytical sample: 1 n = 78 (32 males, 46 females; 33 urban, 45 rural) for all physical activity intensities. One child did not participate in the study; one monitor number was missed to enter, two parent/caregivers denied using the accelerometer on their children, six data were lost during transfer, and 12 children wore the accelerometer for less than 24 hours. Differences between sexes and place of residence were tested using Wilcoxon's rank-sum test 2 n=99 (41 males, 58 females; 49 urban, 50 rural). Differences between sexes and place of residence were tested using the Pearson chi-square test.

Around 80% of the children met the recommendations for MVPA (≥ 60 min/day), 60% for TPA (≥ 180 min/day), 56% for screen time (≤ 60 min/day), 82% for sleep (10–13 h/day) and 37% for restrained sitting (37%). Only one in ten participants met all three recommendations (physical activity, sedentary behaviour and sleep). A significantly higher proportion of females met the daily screen time recommendation than males (65% vs. 42%, p = 0.018). None of the 41 males included in the study met all three recommendations (0% males vs. 17% females, p = 0.013), and no statistically significant differences were observed by place of residence.

Meeting all recommendations: This included physical activity, sedentary behaviour and sleep.

Table 3 shows the results of the executive functions tasks: inhibition and visual-spatial working memory. The average score for inhibition was 0.6 ± 0.2, with females displaying better inhibitory control than males (0.6 ± 0.2 vs 0.5 ± 0.2, p = 0.008). The average score for working memory was 2.7 ± 1.0, with no statistically significant difference between females and males. Rural children showed higher working memory scores than urban children (3.1 ± 0.7 vs 2.4 ± 1.1, p<0.001). Overall, 97% of

the participants were either within or above age expectations for executive functions.

Table 3 Executive function [(Mean ± SD) or n (%)] of participating children, stratified by sex and place of residence Characteristics All (n = 91) Males (n = 39) Females (n = 52) p -value Urban (n = 45) Rural (n = 46) p -value Executive functions (Mean ± SD) Go/No-Go (Inhibition) 0.6 ± 0.2 0.5 ± 0.2 0.6 ± 0.2 0.008* 0.6 ± 0.2 0.6 ± 0.2 0.093 Mr. Ant (Visual-spatial working memory) 2.7 ± 1.0 2.9 ± 1.0 2.6 ± 1.0 0.270 2.4 ± 1.1 3.1 ± 0.7 < 0.001* Go/NoGo developmental status 1 Above age expectations 40 (43.9%) 15 (38.5%) 25 (48.1%) 0.513 21 (46.7%) 19 (41.3%) 0.685 Within age expectations 48 (52.7) 22 (56.4%) 26 (50.0%) 22 (48.9%) 26 (56.5%) Somewhat below age expectations 3 (3.3%) 2 (5.1%) 1 (1.9%) 2 (4.4%) 1 (2.2%) Mr Ant developmental status 2 Above age expectations 74 (81.3%) 33 (84.6%) 41 (78.8%) 0.510 33 (73.3%) 41 (89.1%) 0.053 Within age expectations 17 (18.7%) 6 (15.4%) 11 (21.2%) 12 (26.7%) 5 (10.9%) 1 Go/No-GO: The “GO/No-Go” assessed the behavioural urges and impulses of the child. In this game, fish and sharks were presented, and the child was instructed to tap the iPad screen when fish appeared and not to tap when sharks appeared. The total score was calculated as percentage accuracy (range between 0 and 1, with 1 referring to a perfect score) by multiplying the proportions of go trials (catching the fish) and no-go trials (not catching the shark) 2 Mr. Ant: This game measured the child’s visual-spatial working memory, i.e., the amount of visual information that can be actively maintained. In this game, an ant was presented with several colored dots in different spatial locations on its body, which vanished after some time. Then, the child was asked to recall and tap the locations of the dots. The scoring begins at level 1, and participants earn 1 point for each consecutive level in which at least two of the three trials were accurate. After the initial levels, participants also earn an additional third of a point for each correct trial thereafter. Therefore, the final score could range between 0 and 8 points, * denotes the value is significant

One child did not participate in the study, one refused to play Go/No Go, and the data of seven children on executive functions were missing; therefore, for inhibition and visual-spatial working memory, n = 91 (39 males and 52 females; 45 urban and 46 rural). Differences between sexes and place of residence were tested using the Pearson chi-square test

Table 4 presents the overall scores for the gross and fine motor skills by sex and place of residence. Males displayed higher handgrip strength than females (5.2 ± 1.8 kg vs. 4.3 ± 1.5 kg), p = 0.009. However, females showed better fine motor skills than males, taking less time to complete the 9-hole pegboard test (p = 0.007). Rural children demonstrated better gross motor skills, including standing long jump, one-leg standing balance, and handgrip strength, as well as fine motor skills (pegboard test), than their urban counterparts.

Table 4 Overall scores (Mean ± SD) for the gross and fine motor skills, stratified by sex and place of residence Characteristics All (n = 98) Males (n = 40) Females (n = 58) p -value Urban (n = 49) Rural (n = 49) p -value Gross motor skills S-TUG (s) 8.1 ± 6.8 9.2 ± 9.9 7.3 ± 3.3 0.160 7.8 ± 8.0 8.3 ± 5.5 0.700 Standing long jump (cm) 65.4 ± 21.5 68.5 ± 22.8 63.3 ± 20.5 0.240 56.0 ± 18.1 74.8 ± 20.6 < 0.001* Balance (s) 7.0 ± 11.8 5.4 ± 6.7 8.2 ± 14.2 0.240 3.9 ± 2.4 10.1 ± 16.0 0.008* Handgrip (kg) 4.7 ± 1.7 5.2 ± 1.8 4.3 ± 1.5 0.009* 3.9 ± 1.5 5.5 ± 1.5 <0.001* Fine motor skills Pegboard (s) 40.8 ± 9.5 43.9 ± 8.9 38.7 ± 9.5 0.007* 43.0 ± 9.3 38.7 ± 9.4 0.023* * denotes the value is significant, Abbreviations : S-TUG Standing Timed Up and Go, s seconds, cm centimetre, kg kilogram. Differences between sex and place of residence were tested using Wilcoxon’s rank-sum test One child did not participate in the study, and the other refused to perform the motor skill tests; therefore, n = 98

Table 5 presents the duration of outdoor time, restrained sitting, sedentary screen time and sleep based on parent/caregiver report.

Table 5 Parent/caregiver questionnaire results (mean ± SD), by sex and place of residence Characteristics All (n = 99) Males (n = 41) Females (n = 58) p -value Urban (n = 49) Rural (n -50) p -value Time spent outdoors Time spent outdoors (weekdays, hr/day) 2.9 ± 2.1 2.7 ± 2.1 3.0 ± 2.1 0.500 3.4 ± 2.5 2.4 ± 1.5 0.011* Time spent outdoors (weekends, hr/day) 2.6 ± 1.8 2.6 ± 1.8 2.6 ± 1.8 0.870 2.0 ± 1.6 3.3 ± 1.7 < 0.001* Restrained sitting (i.e. time spent in a vehicle) Time spent sitting in a vehicle (weekdays, hr/day) 1.0 ± 1.0 0.9 ± 0.9 1.0 ± 1.1 0.710 1.1 ± 0.9 0.9 ± 1.1 0.380 Time spent sitting in a vehicle (weekends, hr/day) 0.2 ± 0.6 0.2 ± 0.4 0.3 ± 0.7 0.370 0.2 ± 0.4 0.3 ± 0.7 0.630 Sedentary screen time (hr/day) 1.4 ± 1.4 1.8 ± 1.6 1.2 ± 1.0 0.018* 1.1 ± 1.0 1.7 ± 1.6 0.035* Sleep (h/day) 10.5 ± 1.1 10.4 ± 0.9 10.6 ± 1.3 0.460 10.6 ± 1.3 10.4 ± 1.0 0.320 * denotes the value is significant, Differences between sex and place of residence were tested using Wilcoxon's rank-sum test

Participants, on average, spent around 3 h outdoors on weekdays and slightly less (2.6 h) on weekends. Rural children spend significantly less time outdoors on weekdays but more time outdoors on weekends than urban children. Males than females and rural children than urban children had higher sedentary screen time. No statistically significant difference was observed for restrained sitting and sleep by sex and place of residence.

Acceptability and implementation feasibility of the study protocol

Altogether, 48 parents/caregivers participated in the FGDs. Participants had a mean age of 31 years (SD = 11), and the majority (93%) were female. Regarding educational background, nearly half (48%) had completed secondary education, while only 1% held a university qualification. All the ECD teachers were female, with a mean age of 21 years (SD = 5). Table 6 presents the summary of the FGD and interview findings.

Table 6 Findings of focus group discussion with parents/caregivers and ECD teachers Key themes Main findings Implications for the study protocol Parent/caregiver feedback on questionnaire • Questionnaires were well understood due to the Nepali translation • Estimating PA and screen time was challenging; it required data collector support • Retain Nepali translation • Data collectors should guide the estimation of PA and screen time Children's overall response to participation • Children were highly engaged throughout the study • Confirms acceptability and appropriateness of the study measures Feedback on accelerometer use • Mostly well accepted by parents/caregivers • Two parents/caregivers declined due to safety concerns • Some discomfort reported around device use during sleep • Provide clear information on device safety and benefits • ECD teachers should communicate this, as they are more trusted by families Feedback on executive and motor skill tasks • Children actively participated and appeared happy • Parents/caregivers reported that they eagerly observed and sought individual results • Confirms acceptability and appropriateness of assessment tasks EDC teacher's observation • Data collection techniques were novel and well-received • Teachers actively observed and supported data collectors • Teachers were satisfied with the overall data collection process • Confirms ECD teacher acceptance and endorsement of the study protocol ECD Early Childhood Development, PA Physical activity

Overall, the qualitative findings confirmed the acceptability and feasibility of the SUNRISE study protocol among parents/caregivers and ECD teachers. While the Nepali translation of the questionnaire facilitated comprehension, estimating total physical activity and screen time remained challenging, indicating a need for structured guidance from data collectors in future studies. Children demonstrated high engagement across all executive and motor skill tasks, and the accelerometer was well accepted by most families, though a small number raised concerns about potential harmful

effects or discomfort during sleep, underscoring the importance of clear communication about device safety before data collection. ECD teachers found the process novel and engaging, actively supporting data collectors throughout. Collectively, these findings affirm the protocol's readiness for large-scale implementation, while highlighting targeted areas for refinement.

Discussion

To our knowledge, this is the first study in Nepal to objectively assess 24-hour movement behaviours among children using the WHO Global Guidelines. Overall, only one in ten preschool children met all the WHO-recommended 24-hour movement guidelines. The findings also reported notable urban–rural and sex-related differences in children's adherence to the recommendations and the crucial need for promoting 24-hour movement behaviours in Nepali children. More importantly, this pilot study demonstrated the feasibility of implementing the SUNRISE study protocol in diverse settings (urban/rural) of Nepal

This pilot study reported that only 10% of preschool children met all the recommendations of the WHO guidelines. This prevalence was lower than that reported in several developed countries, including Sweden (19.4%) [27], Australia (14.9%) [28], and Canada (12.7%) [29]. It was also lower than findings from some Asian and South Asian countries, such as China (15%) [30] and Sri Lanka(18%) [31]. However, Nepal's estimate was higher than that of Bangladesh (4.7%) [32]. These variations across countries could be explained by differences in the socio-economic and environmental contexts of the study sites. In Nepal, the pilot study was conducted in a less-densely populated mid-hilly district representing both urban and rural contexts, whereas studies in Bangladesh, Sri Lanka, and China were undertaken in metropolitan areas [30 – 32]. Whatever the context, this study finding highlights the crucial need for promoting 24-hour movement behaviours in Nepali children.

About 60% of Nepali preschool children met the TPA recommendation of at least 180 min per day. This proportion was higher than that reported in neighbouring South Asian countries, such as Sri Lanka (53%) [31] and Bangladesh (41%) [32], where data were collected in metropolitan settings. However, it was substantially lower than in developed countries such as Sweden and Australia, where nearly 90% of children met the TPA recommendation [27 , 28]. These cross-country comparisons are generally valid because all studies followed the SUNRISE protocol and used waist-worn ActiGraph accelerometers, except in Canada [29], which used a waist-worn Actical monitor. The low proportion of TPA in this study could also be attributed to the limited availability of safe playgrounds and parks for children, particularly in urban settings such as Waling municipality. The TPA findings underscore the need to encourage more active time among Nepali preschool children and to create more supportive environments for physical activity. Nevertheless, the results of this pilot study should be interpreted with caution due to the small sample.

Likewise, we found that approximately 56% of the Nepali preschool children met the recommended sedentary screen time guidelines. This proportion was higher than that reported in many developed and developing countries, including Australia (17.3%), Canada (24%), Sweden (38.7%), Iran (19.5%), Sri Lanka (53%), and Bangladesh (17.5%) [27 – 29 , 31 – 33], but lower than in China (88.2%) [30], where the prevalence was exceptionally high. The Chinese study attributed this to cultural factors, although it did not clearly specify what those factors were [30]. The relatively higher adherence to sedentary screen time guidelines in Nepal may be due to lower access to screen devices, such as tablets and iPads, as well as internet availability, Wi-Fi access, and the cost of internet services in the study sites at the time of data collection. However, sedentary screen time among preschool children

may be higher in metropolitan areas such as Kathmandu Valley, where access to devices and internet connectivity is more widespread [34].

This study also revealed notable differences in screen time compliance by sex and place of residence. Females accumulated fewer minutes of screen time per day and were more likely to meet the recommended guidelines than males. Children living in rural areas had, on average, almost 1.5 times higher daily screen time than children from urban areas. A potential explanation could be social desirability bias, as parents/caregivers in urban areas, who tend to have higher education and greater access to information, might have underreported their children's screen time [35]. Another plausible explanation could be that better-educated parents/caregivers may have actively limited their children's screen use [36]. Given the small sample size and the exploratory nature of this study, these findings need to be interpreted cautiously and confirmed in larger and more diverse samples. Overall, our findings highlight the need for evidence-based interventions to reduce screen time among preschoolers [37], given its association with poorer developmental, language, and behavioural outcomes [38 – 42].

Approximately, 82.0% preschool children in Nepal met the sleep recommendations, a proportion comparable to that reported in developed countries like Canada (83.9%), Australia (88.7%), Belgium (94.0%) and Finland (75.7%) [12 , 28 , 43 , 44], but much higher than in several developing countries, including Sri Lanka (65.0%), Bangladesh (29.5%), China (29.5%), and South Africa (66.0%) [30 – [32 , 45]. Most of these countries used similar SUNRISE protocols to measure sleep, except Finland, which collected data for seven days instead of five days. This study was carried out in the less densely populated mid-hill district of Nepal, but the proportion of children meeting sleep guidelines could be lower in metropolitan cities such as Kathmandu, where the traffic noise [46], increased access to screen devices and the internet [34 , 47 , 48], and limited outdoor play spaces [49] could negatively impact sleep. Urban-rural differences in sleep patterns, as well as potential variations between objective and subjective measurements, warrant further investigation.

We also found that Nepali preschool children had comparable fine motor skills to those in other developing countries, such as Bangladesh [32] and Zimbabwe [50], and performed better than children in Sweden, a high-income country [27]. Girls outperformed boys in fine motor skills, consistent with previous studies reporting better motor performance among preschool-aged girls compared to boys [51 – 54]. Rural children also outperformed urban children in fine motor skills; while this is consistent with findings from Zimbabwe [50], the SUNRISE pilot study from Bangladesh reported no difference between urban and less developed areas [32]. Gross motor skills, particularly the standing long-jump, were notably higher among Nepali children, particularly among rural children, compared to SUNRISE pilot study findings from Bangladesh [32], Zimbabwe [50] and Sweden [27]. The superior performance of rural children may reflect greater access to open play spaces and varied terrain in rural settings. Nepali children also scored comparably or better on measures of inhibition and working memory relative to Bangladesh [32], Zimbabwe [50] and Sweden [27]. These findings, however, require confirmation in larger studies. The findings of this pilot study demonstrate the strong acceptability and feasibility of implementing the SUNRISE study protocol in the Nepali ECD setting. The high participation and engagement of children, parents/caregivers, and ECD teachers across both urban and rural centres reflect the cultural appropriateness of the adapted measures. The translation of study materials into Nepali was a critical facilitator of comprehension among parents/caregivers, consistent with evidence that linguistically adapted tools improve participant understanding and data quality in low- and middle-income country (LMIC) settings [55]. However, the difficulty parents/caregivers experienced in estimating physical activity and screen time highlights a commonly reported limitation of self- and proxy-report measures in this age group and

underscores the need for structured interviewer guidance or simplified estimation aids in the main study. The high acceptance of accelerometer use is noteworthy, given that wearable device use among young children remains relatively novel in Nepal; nonetheless, the concerns raised by a small number of families regarding device safety and discomfort during sleep suggest that a structured parental orientation session, facilitated by trusted ECD teachers, should be incorporated into the main study protocol. The active involvement and satisfaction of ECD teachers further affirm the viability of school-based data collection in this context. Collectively, these findings are consistent with the SUNRISE pilot studies conducted in other LMICs, including Bangladesh [32] and Zimbabwe [50], which similarly reported high protocol acceptability, while identifying context-specific adaptations needed for large-scale implementation.

Strengths and limitations

This is the first study in Nepal to objectively assess 24-hour movement behaviours among preschool-aged children using ActiGraph accelerometers, while simultaneously evaluating executive function and gross and fine motor skills within the standardised international SUNRISE protocol. The use of objective measurement tools minimises the recall bias and social desirability biases inherent in proxy-report measures. A balanced sample of males and females from both urban and rural settings enables meaningful subgroup and cross-country comparisons within the SUNRISE network. The high participation rate and successful protocol implementation across diverse settings further affirm the acceptability and feasibility of the SUNRISE protocol for broader application in Nepal.

Several limitations must be acknowledged. The purposive convenience sample of 99 children from a single mid-hill district limits generalisability to other ecological zones, ethnic groups, and urban contexts such as Kathmandu valley. The small sample size, compounded by accelerometer data loss, reduces statistical power to detect modest subgroup differences. Screen time and sleep data relied on proxy reports, introducing susceptibility to recall and social desirability bias. The cross-sectional design precludes causal inference, and the single data collection period does not account for seasonal variation in movement behaviours. Future studies should include larger, more diverse samples across ecological zones, ethnicities, and seasons to generate nationally representative evidence to inform policy and intervention.

Conclusions

This pilot study is the first in Nepal to objectively assess 24-hour movement behaviours among preschool children, reporting that only one in ten met all WHO-recommended guidelines, with notable disparities by sex and place of residence. While sleep adherence was comparatively high, compliance with physical activity and screen time recommendations remains suboptimal and warrants urgent attention. Nepali preschoolers demonstrated relatively better motor skills and executive function compared to SUNRISE findings from other countries, with rural children consistently outperforming urban peers, likely attributable to greater open access to open play spaces. The high acceptability and feasibility of the SUNRISE protocol across diverse settings affirm its readiness for large-scale implementation in Nepal, with targeted refinements recommended around physical activity estimation guidance and accelerometer onboarding. Future studies encompassing broader ecological zones, ethnic diversity, and seasonal variation are needed to generate nationally representative evidence to inform policies and interventions promoting healthy movement behaviours among Nepali preschool children.

Supplementary Information

Supplementary Material 1.